

Question:

A 5-year-old boy is brought to the physician by his mother because he does not “listen to her” anymore. The mother also reports that her son cannot concentrate on any tasks lasting longer than just a few minutes. Teachers at his preschool report that the patient is more active compared to other preschoolers, frequently interrupts or bothers other children, and is very forgetful. Last year the patient was expelled from another preschool for hitting his teacher and his classmates when he did not get what he wanted and for being disruptive during classes. He was born at term via vaginal delivery and has been healthy except for 3 episodes of acute otitis media at the age of 2 years. He has met all developmental milestones. His mother has major depressive disorder and his father has Graves' disease. He appears healthy and well nourished. Examination shows that the patient does not seem to listen when spoken to directly. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in treatment?

Answer Choices:

- A. Behavior therapy
- B. Methimazole
- C. Fluoxetine
- D. Hearing aids

Reasoning + Answer:

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Finding Reasoning Paths :

1. **ADHD as a potential diagnosis**: The child exhibits symptoms of hyperactivity (disruption in class , forgetfulness , inability to focus), impulsivity (inability to wait for turn , acting without thinking), and inattention (difficulty following instructions , forgetfulness). These symptoms are consistent with attention-deficit/hyperactivity disorder (ADHD).

2. **Family history and socioeconomic factors**: The mother has major depressive disorder , and the father has Graves ' disease . There is also a history of the mother abusing medication , which may indicate socioeconomic stressors or environmental factors that could influence the child 's behavior .

3. **Medication-based treatment for ADHD**: If ADHD is confirmed , methylphenidate is commonly used as a first-line treatment for ADHD in children .

Reasoning Process :

1. **Symptoms Analysis** :

- **Hyperactivity**: The child exhibits hyperactivity and impulsivity , as seen in preschoolers , which aligns with a higher prevalence of ADHD in this age group .

- **Inattention**: Difficulty following instructions , forgetting to bring things , and inability to sit still are hallmark symptoms of inattention .

- **Impulsivity**: Frequent interruptions , talking without thinking , and forgetting what someone is saying suggest impulsivity .

2. **Exclusion of Other Conditions** :

- **Autism spectrum disorder**: While the child shows some signs of inattention and distractibility , there is no mention of repetitive behaviors , social impairments , or restricted interests , which are hallmark features of autism .

- **Mood disorders**: The mother 's major depressive disorder may contribute to the child 's symptoms , but the specific behavioral symptoms (e.g . , hyperactivity , impulsivity , inattention) are more consistent with ADHD .

- **Sensory integration issues**: The absence of specific sensory processing difficulties in the history makes this less likely .

3. **Family History and Socioeconomic Factors** :

- The mother 's major depressive disorder and medication abuse could contribute to the child 's behavioral issues , but these are more indirect factors . The child 's symptoms are more directly aligned with ADHD than with mood disorders or sensory integration issues .

4. **Treatment Considerations** :

- **Behavior therapy**: This is often the first treatment approach for ADHD , especially in young children . However , the question does not provide information about the efficacy of behavior therapy for this child , so methylphenidate may be a more direct pharmacological approach .

- **Methylphenidate**: This is a stimulant used to treat ADHD by increasing dopamine and norepinephrine levels in the brain . It is effective in improving attention , reducing hyperactivity , and impulsivity . It is the most commonly prescribed medication for ADHD in children .

5. **Alternative Medications** :

- Other medications , such as amphetamine-dextroamphetamine (Adderall) or extended-release methylphenidate (Concerta), are also used to treat ADHD . However , methylphenidate is a reasonable and common choice , and the choice of methylphenidate vs . other medications would depend on specific clinical factors not provided in the question .

Conclusion :

The most appropriate next step in treatment for this child with symptoms consistent with ADHD , hyperactivity , impulsivity , and inattention is **methylphenidate** . This is supported by the symptoms , family history , and absence of other underlying conditions .

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<answer>

Methimazole

</answer>

1.5

1.0

0.5

0.0

-0.5

-1.0

-1.5